# DARE TIMOTHY AWOBIRO

08143333645, 09058299750

10 Fagbile St, Lagos, Nigeria https://www.linkedin.com/in/dare-timothy-awobiro dawobiro02@gmail.com

TIMTRIX-DARE 08143333645

## ABOUT

Highly innovative Embedded Systems Engineer with a proven track record in developing advanced robotic systems, IoT security solutions, and intelligent control mechanisms. Expertly leverages C, Python, ESP32, and SolidWorks to design, prototype, and implement complex hardware-software integrations. Adept at leading technical projects, fostering collaborative environments, and delivering impactful presentations, driving significant advancements in mechatronic engineering and student innovation.

## WORK

### Event planner

**Bench cafe** **Apr 2023 - Present, Minna, Niger State, Nigeria**

Organized technical showcases, tutorials, hackathons, and community engagement programs to facilitate knowledge sharing and skill development among engineering students.

- Organized and executed over 15 technical showcases, tutorials, hackathons, and capacity-building initiatives, facilitating knowledge sharing among 200+ engineering students.
- Developed and delivered compelling presentations and reports for key events, effectively highlighting student innovation and engineering projects to increase departmental visibility by 25%.
- Assisted in the planning and execution of numerous community engagement programs and technical workshops, directly impacting local student communities.
- Prepared comprehensive reports, presentations, and documentation, providing critical support for organizational goals and strategic initiatives.

### Embedded Systems Engineer

**FUTminna** **Mar 2023 - Present, Minna, Niger State, Nigeria**

Designed and developed advanced embedded systems for robotics, IoT security, and automation, integrating hardware and software components.

- Developed a fully functional robotic vacuum cleaner, integrating ESP32, proximity sensors, and motor control systems for autonomous navigation and efficient cleaning operations.
- Designed and built an advanced embedded security system leveraging facial recognition technology and microcontrollers for automated door access control, recognized at the Deep Fund Hackathon.
- Engineered a robust sensor-based alarm system capable of real-time motion detection and alert triggering, integrating hardware and software components for enhanced security.
- Programmed and implemented a PID-controlled line-following robot using Arduino, demonstrating expertise in control systems and robotic navigation at Mechatronics Student Association competitions.
- Modeled and simulated a PID-Controlled Mass-Spring-Damper System using MATLAB Simulink, performing comprehensive stability analysis and optimizing PID tuning for enhanced performance.
- Led team-based projects integrating mechanical and electronic systems to deliver practical engineering solutions, addressing real-world challenges.
- Presented project findings in multiple departmental and university-wide technical presentations, showcasing interdisciplinary engineering approaches.

### President

**National association of Mechatronic Engineering** **Apr 2025 - Apr 2026, Minna, Niger State, Nigeria**

Led a student community of 500+ members, organizing technical workshops and innovation challenges while representing the association at university and external forums.

- Spearheaded the growth and engagement of a 500+ member student community, organizing over 10 technical workshops, hackathons, and innovation challenges.
- Represented the association at over 5 university and external technical forums, delivering impactful presentations on student innovation and departmental achievements.
- Developed and implemented strategic initiatives that resulted in a 20% increase in participation in technical competitions and capacity-building programs.
- Fostered collaborative learning environments and facilitated knowledge sharing, contributing to a vibrant mechatronic engineering community.

### Manager/supervisor

**Bench cafe** **Aug 2024 - Apr 2026, Minna, Niger State, Nigeria**

Managed daily operations, staff scheduling, and inventory to ensure seamless workflow and optimal resource utilization for a cafe.

- Managed daily operations, including staff scheduling for a team of 5+ members, inventory control, and resource allocation, ensuring seamless workflow and optimal utilization.
- Trained and mentored new and existing staff, implementing process improvements that enhanced operational efficiency by 15% and significantly improved customer experience.
- Developed and enforced operational standards, resulting in improved service delivery and a measurable increase in team productivity.
- Oversaw financial transactions and daily reporting, contributing to accurate record-keeping and operational transparency.

### Teacher of physics and mathematics

**Semait school** **Jan 2022 - Jun 2025**

Taught Physics and Mathematics to secondary school students, prepared lesson plans, and assisted with exam preparation and problem-solving.

- Taught Physics and Mathematics to secondary school students across multiple grade levels, achieving an average 10% improvement in student test scores.
- Prepared engaging lesson plans and conducted practical demonstrations, enhancing student comprehension of complex scientific and mathematical concepts.
- Assisted over 50 students annually in exam preparation and problem-solving skills development, contributing to their academic success.
- Provided individualized support and tutoring, fostering a positive learning environment and addressing diverse student needs.

### AutoCAD Designer

**Fronthill Limited** **Jan 2025 - Mar 2025**

Created precise digital designs and tested product/system behavior to ensure physical build accuracy and optimal performance.

- Produced precise digital designs and schematics using AutoCAD for various products and systems, ensuring high fidelity for physical construction.
- Conducted advanced simulations and virtual testing to predict and optimize product behavior, identifying and resolving potential issues before physical prototyping.
- Collaborated closely with engineering teams to translate complex conceptual designs into detailed, manufacturable blueprints, improving design-to-production efficiency.
- Contributed critical design insights to the development cycle, ensuring all components met stringent performance and integration standards.

## EDUCATION

### Mechatronics Engineering

• Federal university of technology minna • Bachelor of Engineering (Expected) - Minna, Niger State, Nigeria • Jan 2023

### MECHANICAL ENGINEER

Upper Credit • Federal Polytechnic Bida, Niger State • Ordinary National Diploma - OND - Bida, Niger State, Nigeria • Jan 2020 - Dec 2022

### Science

• decs new college • High School Diploma • Jan 2015 - Dec 2019

## SKILLS

**Embedded Systems & Robotics** | Microcontroller (ESP32), Circuit Assembly, Prototyping, Power Management, PID Control, Circuit Design, Motor Control, Navigation Algorithms, IoT Security, Facial Recognition, Arduino, Real-time Alerts, Sensor-Based Systems, Line Following Robotics

**Programming & Software** | C, Python, MATLAB, MATLAB Simulink, Programmer, Hardware and software troubleshooting

**Design & Graphics** | AutoCAD, AutoCAD Electrical, SolidWorks, Graphics Designer, Autocad Drawing, Motion Graphics

**Project Management & Leadership** | Team Leadership, Systems Integration, Project Management, Strategic Planning, Event Planning, Community Engagement, Operations Management, Staff Management, Technical Presentations

**Education & Training** | Teaching, Curriculum Development, Mentorship, Problem-solving

## CERTIFICATES

### IEE ROBOTIC ENGINEERING
IEEE
Nov 2026

### CAPACITY BUILDING
NUESA
Sep 2025

### MATLAB
MATLAB WORK
Oct 2024

## LANGUAGES

English